Kiran Yamanappa Alur

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Summary

Electronics and Communication Engineering student with a strong interest in digital systems and embedded technologies. Gained hands-on experience in Verilog HDL, FPGA design, embedded systems, C/C++, and Python through academic and internship projects Strong analytical and problem-solving skills, with the ability to collaborate effectively in team environments and adapt quickly to new technologies. Passionate about developing practical engineering solutions and eager to contribute to innovative projects while continuously expanding technical expertise.

Skills

Languages: C, C++, Python | **HDL:** Verilog HDL | **Embedded Systems: Arduino, ESP32**| **Tools:** Vivado, Vitis Model Composer, MATLAB/Simulink, ModelSim/QuestaSim, Git, VS Code | **Concepts:** Digital Electronics, Computer Architecture.

Experience

Summer Intern

Crofting Technologies | *[22 June 2026] – Present*

- Designed and implemented a Verilog HDL-based RISC-V processor integrated with an Ethernet communication interface, performing RTL development, functional simulation, debugging, and FPGA validation using Xilinx Vivado and Vitis Model Composer. Achieved 84% code coverage for the Ethernet module using ModelSim/QuestaSim.
- Developing an FPGA-based object detection system by implementing image processing algorithms, validating them using MATLAB/Simulink, optimizing RTL, and resolving hardware integration issues. Achieved 93% detection accuracy and 78% precision during system evaluation.

Projects

Women's Safety System with Integrated Mobile Application

- Designed and developed a women's safety system capable of sending emergency alerts with the user's live location. Validated the system by conducting integration testing across 20 different locations and 30 user accounts, verifying reliable synchronization between the Android application and web portal while ensuring robust event handling and system reliability. Technologies: Android Studio, HTML, CSS, Node.js

Air Mouse Using ESP32

- Developed a wireless air mouse using ESP32 and the MPU6050 motion sensor. Implemented Bluetooth Low Energy (BLE) communication to control the computer cursor without additional drivers.
- Built a prototype currently used to assist individuals with paralysis and children with physical disabilities in computer interaction, demonstrating the potential of low-cost assistive technology.
- Processed accelerometer and gyroscope data to achieve smooth cursor movement and gesture-based control, improving responsiveness through iterative testing and calibration.

Certifications

Programming in C – Wingspan (Aug 2024): Completed practical training in C programming, including control structures, arrays, pointers, functions, and structures.

Basics of Python – Wingspan (Nov 2023): Completed foundational training in Python, covering variables, data types, loops, functions, and basic programming **concepts**.

Education

Qualification	Institution	Year of passing	Score
B.E. (ECE)	SDM College of Engineering & Technology, Dharwad	2027*	9.38(up to 6th Semester)
PUC (Science)	Empower PU Science College, Dharwad	2023	91%
SSLC	BGS High School, Dharwad	2021	96%